Finite terms: cancellation of the negative logarithms, reality, and simplification of all rows

1 Notation

All terms of Group I, Row III depend on the transverse coordinates only through

$$a \equiv x - w, \qquad b \equiv x - z, \qquad c \equiv z - w = a - b, \qquad A \equiv a^2, \qquad B \equiv b^2, \qquad c^2 = A + B - 2\,a \cdot b, \tag{1.1}$$

and through the two vectors

$$u^i \equiv \frac{(x' - w')^i}{(x' - w')^2}, \qquad v^j \equiv \frac{(y - z)^j}{(y - z)^2}. \tag{1.2}$$

Useful identities:

$$c \cdot a = A - a \cdot b, \qquad c \cdot b = a \cdot b - B, \qquad A - B = c \cdot (a + b). \tag{1.3}$$

All terms of Row III carry the same colour/Wilson-line factor

$$\mathcal{W}_{\rm III} \equiv \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\rho^{b'}(x')\; f^{abd}\, U^{de}(x)\Big[U^{bc}(z) - U^{bc}(y)\Big]\rho^{e}(x)\,\rho^{c}(y), \tag{1.4}$$

which does *not* depend on w, and the same prefactor. I therefore write

$$\frac{d^3 N_{\rm I.III}}{d^3 k} \equiv \frac{d^3(N_1 + N_2 + N_{3a} + N_{3b})}{d^3 k} = \frac{1}{(2\pi)^3}\,\frac{i g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'} e^{-ik\cdot(w'-w)}\; u^i\, v^j\; \mathcal{R}^{ij}\; \mathcal{W}_{\rm III} \;+\; \text{c.c.}, \tag{1.5}$$

and the task is to compute the tensor $\mathcal{R}^{ij} = \mathcal{R}_1^{ij} + \mathcal{R}_2^{ij} + \mathcal{R}_{3a}^{ij} + \mathcal{R}_{3b}^{ij}$.

2 Group I, Row III: step by step

2.1 Step 1: the four pieces in compact notation

Reading off the file (note the prefactor $1/16\pi^5 = \frac{1}{2}\cdot 1/8\pi^5$ in some terms, and the overall minus sign of N_2):

$$\mathcal{R}_1^{ij} = \frac{\delta^{ij}}{2}\,\frac{A + B}{(B - A)^2}\left\{\log\left[\frac{B}{B - A}\right] + \log\left[\frac{A - B}{A}\right]\right\} - \frac{a^i\, b^j}{A\, B}\,\log\left[\frac{-B}{A}\right], \tag{2.1}$$

$$\mathcal{R}_2^{ij} = -\,M^{ij}\,\log\left[\frac{-B}{A - B}\right], \tag{2.2}$$

$$\mathcal{R}_{3a}^{ij} = -\,M^{ij}\,\log\left[\frac{A - B}{A}\right] + \mathcal{A}_0\left[\frac{c^i a^j}{c^2} + \frac{b^i c^j}{c^2} - \frac{(a \cdot b)\, c^i b^j}{c^2\, B}\right], \tag{2.3}$$

$$\mathcal{R}_{3b}^{ij} = -\,\mathcal{A}_0\,\frac{(a \cdot b)\, a^i c^j}{c^2\, A} - \frac{1}{2}\,\log\left[\frac{c^2}{A}\right]\left[\frac{c^i b^j}{c^2 B} - \frac{a^i c^j}{c^2 A} + \frac{\delta^{ij}}{c^2}\right], \tag{2.4}$$

where the *same* tensor appears in N_2 and in the six logarithmic terms of N_{3a}:

$$M^{ij} \equiv \frac{1}{c^2}\left\{\frac{c^i a^j + b^i c^j}{A - B} - \frac{1}{A - B}\left[\frac{B}{A}\,a^i c^j + \frac{A}{B}\,c^i b^j\right] + \delta^{ij}\,\frac{B\,(c \cdot a) - A\,(c \cdot b)}{(A - B)^2}\right\}. \tag{2.5}$$

(Term by term: the six terms of N_2 inside the braces are exactly M^{ij}, and the six logarithmic terms of N_{3a} are exactly $-M^{ij}$.)

2.2 Step 2: the logarithms of $(A-B)$ cancel

Three different logarithms with sign-indefinite arguments appear:

$$L_a \equiv \log\left[\frac{B}{B-A}\right] = \log\left[\frac{-B}{A-B}\right], \qquad L_b \equiv \log\left[\frac{A-B}{A}\right], \qquad L_c \equiv \log\left[\frac{-B}{A}\right]. \tag{2.6}$$

For $A > B$ the argument of L_a is negative and that of L_b positive; for $A < B$ it is the other way round. In both regions exactly one of the two is negative, and

$$L_a + L_b = \log\left[\frac{B}{B-A}\cdot\frac{A-B}{A}\right] = \log\left[\frac{-B}{A}\right] = L_c = \log\left[\frac{B}{A}\right] + i\,\sigma\,\pi, \qquad \sigma = \pm 1. \tag{2.7}$$

The dependence on $\log|A-B|$ drops out identically. The only remnant of the negative arguments is the constant $i\sigma\pi$, with the *same* sign in both regions $A \gtrless B$ (for the principal branch $\log(-|t|) = \log|t| + i\pi$, $\sigma = +1$; for the opposite $i\epsilon$ prescription $\sigma = -1$).

Using (2.7), $\mathcal{R}_2 + \mathcal{R}_{3a}\big|_{\log} = -M^{ij}(L_a + L_b) = -M^{ij}L_c$, so all logarithms of this type collect into

$$\mathcal{R}^{ij}\Big|_{L_c} = \log\left[\frac{-B}{A}\right]\left\{\frac{\delta^{ij}}{2}\,\frac{A+B}{(A-B)^2} - \frac{a^i b^j}{AB} - M^{ij}\right\}. \tag{2.8}$$

2.3 Step 3: the poles at $A = B$ cancel

δ^{ij} part. With (1.3), $B\,(c\cdot a) - A\,(c\cdot b) = 2AB - (a\cdot b)(A+B)$, hence

$$\frac{A+B}{2(A-B)^2} - \frac{2AB-(a\cdot b)(A+B)}{c^2(A-B)^2} = \frac{\frac{1}{2}(A+B)\,(A+B-2a\cdot b) - 2AB + (a\cdot b)(A+B)}{c^2(A-B)^2} = \frac{\frac{1}{2}(A-B)^2}{c^2(A-B)^2} = \frac{1}{2c^2}. \tag{2.9}$$

Non-δ^{ij} part. With $c = a - b$:

$$\begin{aligned} c^i a^j + b^i c^j &= a^i a^j - b^i b^j, \\ \frac{B}{A}\,a^i c^j + \frac{A}{B}\,c^i b^j &= \frac{B}{A}\,a^i a^j - \frac{A}{B}\,b^i b^j + a^i b^j\,\frac{A^2-B^2}{AB}, \end{aligned} \tag{2.10}$$

so that

$$M^{ij}\Big|_{\text{non}-\delta} = \frac{1}{c^2(A-B)}\left\{a^i a^j\,\frac{A-B}{A} + b^i b^j\,\frac{A-B}{B} - a^i b^j\,\frac{(A-B)(A+B)}{AB}\right\} = \frac{1}{c^2}\left\{\frac{a^i a^j}{A} + \frac{b^i b^j}{B} - \frac{(A+B)\,a^i b^j}{AB}\right\}. \tag{2.11}$$

Adding $-a^i b^j/(AB) = -c^2\,a^i b^j/(c^2 AB)$ and using $A + B - c^2 = 2\,a\cdot b$:

$$\boxed{\mathcal{R}^{ij}\Big|_{L_c} = K^{ij}\,\log\left[\frac{-B}{A}\right], \qquad K^{ij} \equiv \frac{1}{c^2}\left\{\frac{\delta^{ij}}{2} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{2\,(a\cdot b)\,a^i b^j}{AB}\right\}} \tag{2.12}$$

All single and double poles at $(x-z)^2 = (x-w)^2$ have disappeared.

K^{ij} is the kernel of the other rows. Writing $c = a - b$ in the bracket of Group I Row IV (identical to the brackets of Group II Row I ($1a$), Group III Row I and Group III Row II),

$$\frac{b^j c^i}{c^2 B} - \frac{a^i c^j}{c^2 A} - \frac{a^i b^j}{AB} + \frac{\delta^{ij}}{2c^2} = K^{ij}. \tag{2.13}$$

In the original variables

$$K^{ij} = \frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} - \frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2} + \frac{\delta^{ij}}{2(z-w)^2}. \tag{2.14}$$

2.4 Step 4: the $\mathcal{A}_0$ terms

Collecting the four $\mathcal{A}_0$ terms of N_{3a} and N_{3b} and using $c = a - b$ and (1.3):

$$\mathcal{R}^{ij}\Big|_{\mathcal{A}_0} = \frac{\mathcal{A}_0}{c^2}\left\{c^i a^j + b^i c^j - (a\cdot b)\left[\frac{c^i b^j}{B} + \frac{a^i c^j}{A}\right]\right\}$$

$$= \mathcal{A}_0\, T^{ij}, \qquad T^{ij} \equiv \frac{1}{c^2}\left\{ \frac{(c\cdot a)}{A}\, a^i a^j + \frac{(c\cdot b)}{B}\, b^i b^j - \frac{(a\cdot b)\,(A-B)}{AB}\, a^i b^j \right\}. \tag{2.15}$$

$\mathcal{A}_0$ itself is real and elementary. With $\Delta \equiv \sqrt{AB-(a\cdot b)^2} = |a\times b|$ (twice the area of the triangle x, z, w), θ_x the angle between a and b and θ_z the angle between b and $b-a$: $\arctan[(a\cdot b)/\Delta] = \frac{\pi}{2} - \theta_x$ and $\arctan[(B - a\cdot b)/\Delta] = \frac{\pi}{2} - \theta_z$, therefore

$$\mathcal{A}_0 = \frac{\pi - \theta_x - \theta_z}{\Delta} = \frac{\theta_w}{\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}, \qquad \cos\theta_w = \frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|}, \quad 0<\theta_w<\pi, \tag{2.16}$$

i.e. the interior angle at w of the triangle (x, z, w) divided by twice its area. It contains no logarithm and no imaginary part.

2.5 Step 5: the $\log[(z-w)^2/(x-w)^2]$ terms

These have a positive argument. With $c = a - b$,

$$\mathcal{R}^{ij}\Big|_{\log c^2} = -\frac{1}{2}\,\log\!\left[\frac{c^2}{A}\right] S^{ij}, \qquad S^{ij} \equiv \frac{1}{c^2}\left\{ \delta^{ij} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{(A+B)\,a^i b^j}{AB} \right\} = K^{ij} + \frac{\delta^{ij}}{2c^2} + \frac{a^i b^j}{AB}. \tag{2.17}$$

2.6 Step 6: result for Row III

$$\boxed{\mathcal{R}^{ij} = K^{ij}\left\{ \log\!\left[\frac{(x-z)^2}{(x-w)^2}\right] + i\,\sigma\,\pi \right\} - \frac{1}{2}\,S^{ij}\,\log\!\left[\frac{(z-w)^2}{(x-w)^2}\right] + \mathcal{A}_0\,T^{ij}} \tag{2.18}$$

to be inserted in (1.5). The 22 terms of N_1, N_2, N_{3a}, N_{3b} reduce to three structures; (2.18) was checked numerically against the original expressions (2000 random configurations, complex principal-branch logarithms, relative deviation $< 10^{-8}$).

3 The negative logarithms and reality

3.1 What cancels and what does not

- The logarithms of the sign-indefinite combination $(x-w)^2 - (x-z)^2$ cancel *completely*, together with all poles $1/[(x-w)^2-(x-z)^2]$ and $1/[(x-w)^2-(x-z)^2]^2$.
- What is left of the negative arguments is a single constant, $\log[-B/A] = \log[B/A] + i\sigma\pi$. It does *not* cancel inside Row III: it multiplies K^{ij}, and no other term of Row III has that tensor structure with a compensating phase.
- Because of the explicit prefactor i, this constant gives a *real* kernel, $i\cdot i\sigma\pi\,K^{ij} = -\sigma\pi\,K^{ij}$:

$$\left.\frac{d^3N_{\mathrm{I.III}}}{d^3k}\right|_{\pi} = -\,\sigma\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'} e^{-ik\cdot(w'-w)}\; u^i\,v^j\,K^{ij}\;\mathcal{W}_{\mathrm{III}} \;+\; \text{c.c.} \tag{3.1}$$

3.2 Reality

Every contribution of Groups I and III has the form $X + \text{c.c.}$ and is therefore real whatever the phase of X; the explicit factors of i are not a problem. Concretely, for commuting (classical) ρ the Wilson-line factor $\mathcal{W}_{\mathrm{III}}$ is real, and (1.5) with (2.18) gives

$$\frac{d^3N_{\mathrm{I.III}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^5}\,\frac{1}{k^+}\int u^i v^j\,\mathcal{W}_{\mathrm{III}}\Big\{ \sin\big[k\cdot(w'-w)\big]\,\mathcal{R}^{ij}_{\mathrm{real}} - \sigma\pi\,\cos\big[k\cdot(w'-w)\big]\,K^{ij}\Big\}, \tag{3.2}$$

with $\mathcal{R}^{ij}_{\mathrm{real}} = K^{ij}\log[B/A] - \frac{1}{2}S^{ij}\log[c^2/A] + \mathcal{A}_0 T^{ij}$. The real logarithms come with sin, the π term with cos; both are real.

For the other groups:

- Group II, Row I ($1a$, $1b$): N_{1b} is the complex conjugate of N_{1a} (primed $\leftrightarrow$ unprimed coordinates, $i \to -i$), so the sum is real.
- Group II, Row I ($2a$–$2f$) and Row IV: no factor of i, all logarithms have positive arguments, and each kernel is symmetric under the exchange (primed $\leftrightarrow$ unprimed) that maps the Wilson-line factor and the phase into their conjugates. Real.
- Group III and the remaining rows of Group I: $i\times$(real kernel) + c.c., all logarithms positive. Real.

So Row III is the only place where a complex kernel occurs, and there it reduces to the constant $i\sigma\pi$ of (3.1).

4 Simplification across the rows: the $K^{ij}\log[(x-z)^2/(x-w)^2]$ terms

Five contributions have the same kernel $u^i v^j K^{ij}\log[B/A]$, the same measure and the same phase $e^{-ik\cdot(w'-w)}$, and all are of the form "+ c.c." (for Group II the conjugate is N_{1b}):

contribution	coefficient of $\dfrac{1}{(2\pi)^3}\dfrac{ig^4}{8\pi^5 k^+}\,u^i v^j K^{ij}\log[B/A]$
Group I, Row III (from (2.18))	$+1$
Group I, Row IV	$+1$
Group II, Row I, $1a$	$-\frac{1}{2}$
Group III, Row I	-1
Group III, Row II	-1

Define

$$V^{be}\equiv U^{cb}(z)\,U^{ce}(y),\qquad R^{ab}\equiv f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x). \tag{4.1}$$

Then (keeping the operator ordering of the ρ's exactly as in the file, and using only $U^{cb}(z)U^{cg}(z)=\delta^{bg}$):

$$\begin{aligned}
\text{I.IV}: &\quad \big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\,V^{be}\rho^e(y)\,R^{ab},\\
\text{II.I.1}a+\text{III.I}+\text{III.II}: &\quad -\tfrac{1}{2}\Big\{U^{ac'}(w')\{\rho^{c'}(x'),\rho^b(y)\}+U^{ab'}(x')V^{bd'}\{\rho^{b'}(x'),\rho^{d'}(y)\}\\
&\qquad -2\,U^{ac'}(w')V^{be'}\rho^{c'}(x')\rho^{e'}(y)-2\,U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\Big\}R^{ab},\\
\text{I.III}: &\quad \big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\,f^{abd}U^{de}(x)\big[U^{bc}(z)-U^{bc}(y)\big]\rho^e(x)\rho^c(y).
\end{aligned}\tag{4.2}$$

Adding them:

- all terms with $U(w')\,V$ cancel exactly; the terms with $U(x')\,V$ leave only a commutator;
- the $U(z)$ term of Row III cancels against the $U(z)U(x)$ term of $R^{ab}\rho^b(y)$ up to a commutator.

The result is

$$\sum\;=\;\frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,u^i v^j\,K^{ij}\,\log\!\left[\frac{(x-z)^2}{(x-w)^2}\right]\Big(\Omega_0+\Omega_{\rm comm}\Big)+\text{c.c.}, \tag{4.3}$$

$$\Omega_0=\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\Big\{f^{cbd}\,U^{ac}(w)\,\rho^b(y)\rho^d(x)-f^{abd}\,U^{bc}(y)\,U^{de}(x)\,\rho^e(x)\rho^c(y)\Big\}, \tag{4.4}$$

$$\begin{aligned}
\Omega_{\rm comm}=&\;\tfrac{1}{2}\,U^{ab'}(x')\,V^{bd'}\big[\rho^{b'}(x'),\rho^{d'}(y)\big]R^{ab}-\Big(U^{ab'}(x')-\tfrac{1}{2}U^{ab'}(w')\Big)\big[\rho^{b'}(x'),\rho^b(y)\big]R^{ab}\\
&-\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\,f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\big[\rho^b(y),\rho^e(x)\big].
\end{aligned}\tag{4.5}$$

Ω_0 no longer contains $U(z)$: in this part z appears only in the kernel and can be integrated, $H^i(x-w,\,y-x)\equiv\int_z v^j K^{ij}\log[B/A]$. For commuting ρ ($\Omega_{\rm comm}\to 0$) the colour structure of Ω_0 is that of Group I, Row II:

$$\Omega_0\;\xrightarrow{[\rho,\rho]\to 0}\;-\tfrac{1}{2}\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\Big[f^{hbc}\,U^{ah}(w)\{\rho^b(x),\rho^c(y)\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big]. \tag{4.6}$$

(The colour algebra was checked numerically with random orthogonal matrices and a random antisymmetric f.) The natural next step is therefore to compare H^i with the Fourier transform of $R^i(\mathbf{k},\mathbf{p})/(\mathbf{p}^2\mathbf{k}^2)$ of Row II; this comparison has not been done here.

5 Building blocks for the remaining rows

The remaining rows can all be written with three objects. For a pair of transverse vectors (a,b) with $c\equiv a-b$ (vectors a,b,c are not to be confused with the colour indices in U^{ab}, R^{ab}) define

$$E^{km}(a,b)\equiv a^k\left[\frac{c^m}{c^2}+\frac{b^m}{2\,b^2}\right],\qquad G^{km}(a,b)\equiv\left[\frac{c^k}{c^2}-\frac{a^k}{2\,a^2}\right]b^m,\qquad \phi(a,b)\equiv\frac{b^2-a^2}{2\,c^2}, \tag{5.1}$$

with traces

$$e\equiv E^{kk}=\frac{a\cdot c}{c^2}+\frac{a\cdot b}{2\,b^2},\qquad g\equiv G^{kk}=\frac{b\cdot c}{c^2}-\frac{a\cdot b}{2\,a^2}. \tag{5.2}$$

For two tensors $M\cdot N\equiv M^{km}N^{km}$, and $(M^T)^{km}\equiv M^{mk}$. Further

$$n^i(r)\equiv\frac{r^i}{r^2},\qquad R^{ab}\equiv f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x), \tag{5.3}$$

and R'^{ab} is the same object with $(w,x)\to(w',x')$ (same z).

6 Group II, Row I, second part ($2a$–$2f$)

6.1 Step 1: notation and colour factor

$$a = x - w, \quad b = x - z, \quad c = z - w = a - b, \qquad a' = x' - w', \quad b' = x' - z, \quad c' = z - w' = a' - b',$$
$$A = a^2,\ B = b^2,\ A' = a'^2,\ B' = b'^2, \qquad \ell \equiv \log\frac{B}{A}, \quad \ell' \equiv \log\frac{B'}{A'}, \qquad D \equiv A'B - AB'. \tag{6.1}$$

A prime on E, G, ϕ, e, g means evaluation at (a', b'). All six pieces carry the same four-term colour bracket. Using $U^{c'b}(z)U^{cb}(z) = \delta^{c'c}$ and $f^{acd'}f^{acd} = N_c\delta^{d'd}$ it is a product:

$$f^{c'bd'}f^{cbd}\,U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - \cdots + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x) \;=\; R'^{ab}\,R^{ab}. \tag{6.2}$$

(Checked numerically for $SU(2)$.) So

$$\frac{d^3N_{2a+\cdots+2f}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}_{\rm II}\;R'^{ab}R^{ab}, \tag{6.3}$$

and the task is to compute the scalar kernel $\mathcal{K}_{\rm II}$.

6.2 Step 2: the five brackets are products of the building blocks

Reading off the square brackets of $2a, \ldots, 2e$ and comparing with (5.1):

$$P \equiv [\ldots]_{2a} = E\cdot G' + G\cdot E', \qquad X \equiv [\ldots]_{2b} = E\cdot E', \qquad Y \equiv [\ldots]_{2c} = G\cdot G',$$
$$Z \equiv [\ldots]_{2d} = \phi'\,e + \phi\,e', \qquad V \equiv [\ldots]_{2e} = \phi'\,g + \phi\,g', \qquad \frac{d_\perp}{4}\frac{(B-A)(B'-A')}{c^2c'^2} = d_\perp\,\phi\,\phi'. \tag{6.4}$$

For example $E\cdot E' = (a\cdot a')\big[\frac{c}{c^2} + \frac{b}{2B}\big]\cdot\big[\frac{c'}{c'^2} + \frac{b'}{2B'}\big]$ gives the four terms of $2b$, and $E\cdot G'$ gives terms 1–4 of $2a$, $G\cdot E'$ terms 5–8.

6.3 Step 3: coefficients of the logarithms

In units of the prefactor of (6.3):

$$2a:\ \frac{P}{D}\,(\ell - \ell'), \qquad 2d:\ \frac{Z}{D}\left[\frac{B}{B-A}\,\ell - \frac{B'}{B'-A'}\,\ell'\right],$$
$$2b:\ \frac{X}{D}\left[-\frac{B}{A}\,\ell + \frac{B'}{A'}\,\ell'\right], \qquad 2e:\ \frac{V}{D}\left[-\frac{A}{B-A}\,\ell + \frac{A'}{B'-A'}\,\ell'\right],$$
$$2c:\ \frac{Y}{D}\left[-\frac{A}{B}\,\ell + \frac{A'}{B'}\,\ell'\right], \qquad 2f:\ -d_\perp\phi\phi'\left\{\frac{1}{D}\left[\frac{AB\,\ell}{(B-A)^2} - \frac{A'B'\,\ell'}{(B'-A')^2}\right] + \frac{1}{(B-A)(B'-A')}\right\}. \tag{6.5}$$

6.4 Step 4: each combination of logarithms is one integral

With $\Delta(\eta) \equiv (\eta B + A)(\eta B' + A')$, partial fractions give

$$2a:\quad \int_0^\infty \frac{d\eta}{\Delta} = \frac{\ell - \ell'}{D},$$
$$2b:\quad \int_0^\infty \frac{d\eta}{\eta}\left[\frac{1}{\Delta} - \frac{\theta(1-\eta)}{AA'}\right] = \frac{1}{D}\left[-\frac{B}{A}\ell + \frac{B'}{A'}\ell'\right],$$
$$2c:\quad \int_0^\infty d\eta\left[\frac{\eta}{\Delta} - \frac{\theta(\eta-1)}{BB'\,\eta}\right] = \frac{1}{D}\left[-\frac{A}{B}\ell + \frac{A'}{B'}\ell'\right],$$
$$2d:\quad \int_0^\infty \frac{d\eta}{(1+\eta)\,\Delta} = \frac{1}{D}\left[\frac{B\,\ell}{B-A} - \frac{B'\,\ell'}{B'-A'}\right],$$
$$2e:\quad \int_0^\infty \frac{\eta\,d\eta}{(1+\eta)\,\Delta} = \frac{1}{D}\left[-\frac{A\,\ell}{B-A} + \frac{A'\,\ell'}{B'-A'}\right],$$
$$2f:\quad \int_0^\infty \frac{\eta\,d\eta}{(1+\eta)^2\,\Delta} = -\frac{1}{D}\left[\frac{AB\,\ell}{(B-A)^2} - \frac{A'B'\,\ell'}{(B'-A')^2}\right] - \frac{1}{(B-A)(B'-A')}. \tag{6.6}$$

These are exactly the six combinations of Step 3. The two subtractions are the endpoint subtractions of the soft limit $\eta \to 0$ (the $\Lambda \to 0$ side) and of $\eta \to \infty$ (the $\vee \to \infty$ side).

6.5 Step 5: result — the six pieces are a perfect square

Adding the six pieces,

$$\mathcal{K}_{\rm II} = \int_0^\infty d\eta\, \frac{\mathcal{N}(\eta)}{(\eta B + A)(\eta B' + A')}\bigg|_{\rm subtr.}, \qquad \mathcal{N}(\eta) = \frac{X}{\eta} + P + \eta\, Y + \frac{Z + \eta V}{1+\eta} + \frac{d_\perp\, \phi\phi'\, \eta}{(1+\eta)^2}, \tag{6.7}$$

and with Step 2 the numerator factorises:

$$\boxed{\mathcal{N}(\eta) = \Phi^{km}(a,b;\eta)\, \Phi^{km}(a',b';\eta), \qquad \Phi^{km}(a,b;\eta) \equiv \frac{1}{\sqrt{\eta}}\, E^{km} + \sqrt{\eta}\, G^{km} + \frac{\sqrt{\eta}}{1+\eta}\, \phi\, \delta^{km}} \tag{6.8}$$

($\delta^{kk} = d_\perp$). Explicitly, in the original variables,

$$\Phi^{km} = \frac{(x-w)^k}{\sqrt{\eta}} \left[\frac{(z-w)^m}{(z-w)^2} + \frac{(x-z)^m}{2(x-z)^2}\right] + \sqrt{\eta} \left[\frac{(z-w)^k}{(z-w)^2} - \frac{(x-w)^k}{2(x-w)^2}\right] (x-z)^m + \frac{\sqrt{\eta}}{1+\eta}\, \frac{(x-z)^2 - (x-w)^2}{2\,(z-w)^2}\, \delta^{km}. \tag{6.9}$$

So the whole second part is

$$\boxed{\frac{d^3 N_{2a+\cdots+2f}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)} \int_0^\infty d\eta\, \left[\frac{\Phi'^{km}\, R'^{ab}}{\eta B' + A'}\right] \left[\frac{\Phi^{km}\, R^{ab}}{\eta B + A}\right]\bigg|_{\rm subtr.}} \tag{6.10}$$

where "subtr." means: in the $E\cdot E'/\eta$ term replace $1/\Delta \to 1/\Delta - \theta(1-\eta)/(AA')$, and in the $\eta\, G\cdot G'$ term replace $\eta/\Delta \to \eta/\Delta - \theta(\eta-1)/(BB'\eta)$.

Closed form. Doing the η integral (the coefficient of each logarithm is the residue of the integrand),

$$\boxed{\mathcal{K}_{\rm II} = \frac{1}{D}\left\{\mathcal{N}\big(-\tfrac{A}{B}\big)\log\frac{B}{A} - \mathcal{N}\Big(-\tfrac{A'}{B'}\Big)\log\frac{B'}{A'}\right\} - \frac{d_\perp}{4\,c^2 c'^2}} \tag{6.11}$$

$$\begin{aligned}
\mathcal{N}\big(-\tfrac{A}{B}\big) &= P - \frac{B}{A}X - \frac{A}{B}Y + \frac{BZ - AV}{B-A} - \frac{d_\perp\, \phi\phi'\, AB}{(B-A)^2}, \\
\mathcal{N}\Big(-\tfrac{A'}{B'}\Big) &= P - \frac{B'}{A'}X - \frac{A'}{B'}Y + \frac{B'Z - A'V}{B'-A'} - \frac{d_\perp\, \phi\phi'\, A'B'}{(B'-A')^2}.
\end{aligned} \tag{6.12}$$

Both forms were checked numerically against the sum of the original $2a$–$2f$ (3000 random configurations, $d_\perp = 2$ and $d_\perp \neq 2$, relative deviation $< 10^{-10}$; the η integral was also done numerically).

6.6 Consequences

- **No spurious poles.** At $D = 0$ one has $A/B = A'/B'$, so the two residues and the two logarithms coincide and the $1/D$ cancels. At $B = A$ the apparent poles $1/(B-A)$, $1/(B-A)^2$ are compensated by $\ell \simeq (B-A)/A$ and $\phi \propto (B-A)$; likewise at $B' = A'$.
- **Reality.** $\mathcal{K}_{\rm II}$ is real and symmetric under (primed $\leftrightarrow$ unprimed); $R'^{ab}R^{ab}$ and the phase go into their conjugates. For commuting ρ the unsubtracted integrand is $|\mathcal{M}|^2$ with $\mathcal{M}^{km,ab}(z,\eta) = \int_{w,x} e^{ik\cdot w}\, \Phi^{km} R^{ab}/(\eta B + A)$, i.e. manifestly real and non-negative; only the two endpoint subtractions are not of this form.
- **All logarithms have positive arguments.** Nothing like the $i\sigma\pi$ of Group I Row III occurs here.

7 Group II: Row I first part and Row IV

7.1 Row I, first part

With K^{ij} of (2.12), $u^i = n^i(x'-w')$, $v^j = n^j(y-z)$,

$$\begin{aligned}
\frac{d^3 N_{1a}}{d^3 k} = -\frac{1}{(2\pi)^3}\, \frac{ig^4}{16\pi^5}\, \frac{1}{k^+} \int e^{-ik\cdot(w'-w)}\, u^i v^j K^{ij}\, \log\frac{(x-z)^2}{(x-w)^2} \\
\times \Big[U^{ac'}(w')\{\rho^{c'}(x'), \rho^b(y)\} - U^{c'b}(z) U^{ad'}(x') U^{c'e'}(y) \{\rho^{d'}(x'), \rho^{e'}(y)\}\Big] R^{ab},
\end{aligned} \tag{7.1}$$

and $N_{1b} = (N_{1a})^*$. These are the terms already combined in Section 4.

7.2 Row IV: the ξ integral

With $r \equiv y' - y$, $s \equiv k \cdot r$ and $t = 1/\xi$,

$$\mathcal{H}(s) \equiv \int_0^1 d\xi\, \frac{1-\xi}{\xi}\, e^{-is/\xi} = \int_1^\infty dt \left(\frac{1}{t} - \frac{1}{t^2}\right) e^{-ist} = (1+is)\, E_1(is) - e^{-is}, \tag{7.2}$$

where $E_1(z) = \int_1^\infty dt\, e^{-zt}/t$. For $s > 0$,

$$\operatorname{Re}\mathcal{H} = -\mathrm{Ci}(s) - s\left[\mathrm{Si}(s) - \tfrac{\pi}{2}\right] - \cos s, \qquad \operatorname{Im}\mathcal{H} = \mathrm{Si}(s) - \tfrac{\pi}{2} - s\,\mathrm{Ci}(s) + \sin s, \qquad \mathcal{H}(-s) = \mathcal{H}(s)^*. \tag{7.3}$$

The bracket of Row IV is the reflection tensor $\Pi^{ij}(r) \equiv \delta^{ij} - 2r^i r^j/r^2$, so

$$\frac{d^3 N_{\mathrm{II.IV}}}{d^3k} = \frac{1}{(2\pi)^3}\, \frac{g^4 N_c}{8\pi^4}\, \frac{1}{k^+} \int_{x,y,x',y'} \mathcal{H}\big(k\cdot(y'-y)\big)\; n^i(x-y)\, \Pi^{ij}(y'-y)\, n^j(x'-y')\; \mathcal{D}^c(y',x')\, \mathcal{D}^c(y,x), \tag{7.4}$$

with $\mathcal{D}^c(y,x) \equiv [U^{ce}(y) - U^{ce}(x)]\rho^e(x)$. The kernel is symmetric under $(x,y) \leftrightarrow (x',y')$ while $s \to -s$; for commuting ρ only $\operatorname{Re}\mathcal{H}$ survives, so the result is real. $\mathcal{H} \simeq -\log|s|$ at $s \to 0$, which is integrable.

8 Group III

8.1 Row III (first part) + Row V + Row VI

These three have identical kernels, measures and phases; the prefactors are $ig^4/8\pi^4$, $ig^4/4\pi^4$, $ig^4/4\pi^4$. Let

$$\mathcal{D}_q^{c'} \equiv \big[U^{c'e'}(q) - U^{c'e'}(x')\big]\rho^{e'}(x'), \qquad q = y' \text{ or } z. \tag{8.1}$$

Adding the three colour factors (in units of $ig^4/8\pi^4$) and keeping the operator ordering:

$$\begin{aligned}\mathcal{C}_3(q) = {} & 2\, f^{c'd'a}\, \mathcal{D}_q^{c'}\, \big[U^{d'b}(z) - U^{d'b}(x)\big]\rho^b(x)\, \big[U^{ac}(w) - U^{ac}(y)\big]\rho^c(y) \\ & + f^{c'd'a}\, \mathcal{D}_q^{c'} \Big\{U^{ac}(w)U^{d'b}(z) - 2\,U^{ac}(y)U^{d'b}(z) + U^{ac}(y)U^{d'b}(x)\Big\}\big[\rho^c(y), \rho^b(x)\big].\end{aligned} \tag{8.2}$$

For commuting ρ only the first line remains: three LO factors $[U(\text{gluon}) - U(\text{source})]\rho(\text{source})$ tied by one f, which match the three Weizsäcker–Williams factors $n(x'-y')$, $n(x-z)$, $n(y-w)$ of the kernel. (Checked numerically for $SU(2)$.)

The p^+ integrals contain only the phase. With $\eta = p^+/k^+$, $s \equiv k\cdot(y'-z)$, $\vee \to \infty$, and the plus prescription read as $\int d\eta\, [1/(\eta(1+\eta))]_+ f(\eta) = \int d\eta\, [f(\eta) - f(0)]/(\eta(1+\eta))$:

$$\begin{aligned}\int_0^\infty \frac{d\eta}{1+\eta}\, e^{-i\eta s} &= e^{is} E_1(is) \equiv J_0(s), \\ \int_0^\infty d\eta \left(\left[\frac{1}{\eta(1+\eta)}\right]_+ + \frac{1}{1+\eta}\right) e^{-i\eta s} &= -\gamma_E - \log(is) \\ &= -\gamma_E - \log|s| - \tfrac{i\pi}{2}\,\mathrm{sgn}(s), \\ \int_0^\infty d\eta\, e^{-i\eta s} &= \frac{-i}{s - i0}.\end{aligned} \tag{8.3}$$

Hence

$$\begin{aligned}\frac{d^3 N_{\mathrm{III.III(1)+V+VI}}}{d^3k} = {} & \frac{1}{(2\pi)^3}\, \frac{ig^4}{16\pi^5}\, \frac{1}{k^+} \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)} \Bigg\{ \\ & \Big[J_0(s)\; n(y'-z)\cdot n(x'-y')\;\; n(y-w)\cdot n(x-z) \\ & \quad + \big(\gamma_E + \log(is)\big)\; n(y'-z)\cdot n(x-z)\;\; n(y-w)\cdot n(x'-y')\Big]\mathcal{C}_3(y') \\ & + \frac{i}{s-i0}\; n(y'-z)\cdot n(y-w)\; \Big[n(x-z)\cdot n(x'-y')\, \mathcal{C}_3(y') - n(x-z)\cdot n(x'-z)\, \mathcal{C}_3(z)\Big]\Bigg\} + \text{c.c.}\end{aligned} \tag{8.4}$$

8.2 Row III, second part

Here $a = x - w$, $b = x - z$, $c = z - w$ as in Section 1, the right colour factor is R^{ab}, and the left one is $\mathcal{L}_q^{ab} \equiv f^{c'd'a}\, U^{d'b}(z)\, \mathcal{D}_q^{c'}$. The long bracket of the first term is, with $\eta = p^+/k^+$,

$$k^+ [\dots]^{k'm} \equiv \mathfrak{B}^{k'm}(\eta) = \delta^{k'm}\left[\frac{e + \eta\, g}{1+\eta} - \phi + \frac{d_\perp\, \eta\, \phi}{(1+\eta)^2}\right] - \omega(\eta)\, E^{k'm} - G^{k'm} - E^{mk'},$$

$$\omega(\eta) \equiv \left[\frac{1}{\eta(1+\eta)}\right]_+ + \frac{1}{1+\eta}, \tag{8.5}$$

and the short brackets of the second and third terms are both $G^{mk'}$. Therefore

$$\begin{aligned}\frac{d^3N_2}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\int_0^\infty d\eta\;\frac{e^{-i\eta s}}{\eta B + A}\;n^m(y'-z)\Big\{n^{k'}(x'-y')\,\mathfrak{B}^{k'm}(\eta)\,\mathcal{L}^{ab}_{y'} \\ - \eta\,G^{mk'}\big[n^{k'}(x'-y')\,\mathcal{L}^{ab}_{y'} - n^{k'}(x'-z)\,\mathcal{L}^{ab}_{z}\big]\Big\}R^{ab} \\ + \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\;\frac{A}{B\,(B-A)}\;\log\frac{B}{A}\;\;n^m(w'-z)\,G^{mk'}\,n^{k'}(x'-z)\;\mathcal{L}^{ab}_{z}\,R^{ab}\;+\text{c.c.}\end{aligned} \tag{8.6}$$

In the last term the p^+ integral has been done, $\int_0^\infty d\eta\,A/[B(1+\eta)(\eta B+A)] = A\log(B/A)/[B(B-A)]$ (regular at $B=A$), and $n^m(w'-z)\,G^{mk'}\,n^{k'}(x'-z) = \big[n(w'-z)\cdot\big(\frac{c}{c^2}-\frac{a}{2A}\big)\big]\big[b\cdot n(x'-z)\big]$. The remaining η integrals are elementary if wanted: $\int_0^\infty d\eta\,e^{-i\eta s}/(\eta B+A) = B^{-1}e^{isA/B}E_1(isA/B)$.

9 Group I, Row V

9.1 First part

With $\sigma \equiv k\cdot(z-x)$, $\xi = p^+/k^+$ and $[k^+/p^+]_+ f \equiv [f(\xi)-f(0)]/\xi$,

$$\begin{aligned}&\int_0^1 d\xi\,e^{-i\xi\sigma} = \frac{1-e^{-i\sigma}}{i\sigma} \equiv h_0(\sigma), \qquad \int_0^1 d\xi\left[\frac{1}{\xi}\right]_+ e^{\mp i\xi\sigma} = -\mathrm{Ein}(\pm i\sigma), \\ &\mathrm{Ein}(i\sigma) = \gamma_E + \log\sigma - \mathrm{Ci}(\sigma) + i\,\mathrm{Si}(\sigma)\quad(\sigma>0),\end{aligned} \tag{9.1}$$

and $\mathrm{Ein}(-i\sigma) = \mathrm{Ein}(i\sigma)^*$. Then

$$\begin{aligned}\frac{d^3N_1}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\int_{x,z,y,x',y',w'}\Big\{e^{-ik\cdot(w'-z)}\Big[h_0(\sigma)\;n(x'-w')\cdot n(z-x)\;\;n(y-z)\cdot n(y'-x) \\ + \mathrm{Ein}(i\sigma)\;n(y'-x)\cdot n(z-x)\;\;n(x'-w')\cdot n(y-z)\Big] \\ + e^{-ik\cdot(w'-x)}\,\mathrm{Ein}(-i\sigma)\;n(x'-w')\cdot n(y'-x)\;\;n(z-x)\cdot n(y-z)\Big\}\;\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x') \\ \times\Big[f^{adc}\,U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\Big] + \text{c.c.}\end{aligned} \tag{9.2}$$

9.2 Second part: $N_{2a} + N_{2b\,\mathrm{rest}}$

Use the building blocks at $\bar a \equiv y-x$, $\bar b \equiv y-z$, $\bar c = \bar a - \bar b = z-x$ (bars on E, G, ϕ, e, g mean this argument). The long bracket of $N_{2b\,\mathrm{rest}}$ is

$$\begin{aligned}\mathfrak{B}^{im}_V(\xi) = \delta^{im}\Big[\bar\phi\,\big(1-d_\perp\,\xi(1-\xi)\big) - \xi\,\bar e - (1-\xi)\,\bar g\Big] \\ + \left(\left[\frac{1}{1-\xi}\right]_+ - 1\right)\bar E^{im} + \left(\left[\frac{1}{\xi}\right]_+ - 1\right)\bar G^{mi} + \bar E^{mi} + \bar G^{im}.\end{aligned} \tag{9.3}$$

The two integrals of N_{2a} add up to the *same* bracket: the first lacks only the $\bar E^{im}$ term, and the second, after $q^+ = k^+ - p^+$ (which maps its phase and denominator onto those of the first, and reading $\int_{k^+}^0 dq^+ = -\int_0^{k^+} dq^+$ literally), supplies exactly $([1/(1-\xi)]_+ - 1)\bar E^{im}$. The colour factors then add, $\big[-ffU(x)U(z) + N_cU(x)\big] + N_c\big[U(y)-U(x)\big]$, and the $U(x)$ terms cancel:

$$\begin{aligned}\frac{d^3(N_{2a}+N_{2b\,\mathrm{rest}})}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_0^1 d\xi\int_{x,y,z,x',w'}\Bigg\{e^{-ik\cdot(w'-z)}\,e^{-i\xi\,k\cdot(z-x)}\;\frac{n^i(x'-w')\,\mathfrak{B}^{im}_V(\xi)\,n^m(z-x)}{\xi\,(y-x)^2 + (1-\xi)\,(y-z)^2} \\ \times\big[U^{ab'}(x') - U^{ab'}(w')\big]\Big[N_c\,U^{af}(y) - f^{adc}f^{fbe}\,U^{db}(x)\,U^{ce}(z)\Big] \\ - e^{-ik\cdot(w'-x)}\,\big[\xi(1-\xi)-2\big]\;\frac{n(x'-w')\cdot(y-x)\;\;n(y-z)\cdot n(y-x)}{(x-z)^2} \\ \times N_c\big[U^{ab'}(x') - U^{ab'}(w')\big]\big[U^{af}(y) - U^{af}(x)\big]\Bigg\}\rho^{b'}(x')\rho^f(y) + \text{c.c.}\end{aligned} \tag{9.4}$$

The colour factor of the first term is the physical one for a gluon at y splitting into gluons at x and z; at $z \to x$ it becomes $N_c[U(y) - U(x)]$ (because $f^{adc} f^{fbe} U^{db}(x) U^{ce}(x) = N_c U^{af}(x)$), which is the colour factor of the subtraction in the last line, and $\xi(1-\xi) - 2$ is the plus-prescription form of the splitting function $\frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi)$ found before.

10 Summary of the simplified finite term

piece	simplified form
I.II	unchanged (momentum space)
I.III	Eq. (2.18): $K\,[\log(B/A) + i\sigma\pi] - \frac{1}{2} S \log(c^2/A) + \mathcal{A}_0 T$
I.III, I.IV, II.I($1a$,$1b$), III.I, III.II: $K \log(B/A)$ parts	one term, Section 4
I.V first part	ξ integral done, Section 9.1
I.V second part	one kernel $\mathfrak{B}_V$, colour $N_c U(y) - ffU(x)U(z)$, Section 9.2
II.I second part ($2a$–$2f$)	perfect square, Eq. (6.8), Section 6
II.IV	ξ integral done, Section 7.2
III.III(first), III.V, III.VI	one term, three LO factors, Section 8.1
III.III second part	kernel $\mathfrak{B}$, last term integrated, Section 8.2

11 Points to check

1. **Behaviour at $w \to z$.** $\mathcal{W}_{\rm III}$ does not depend on w, and for $c = z - w \to 0$

$$\mathcal{R}^{ij} \;\longrightarrow\; -\frac{\delta^{ij}}{2\,(z-w)^2} \left\{ \log\left[\frac{(z-w)^2}{(x-w)^2}\right] - i\sigma\pi \right\} + \mathcal{O}\left(\frac{1}{|z-w|}\right), \tag{11.1}$$

   because $K^{ij} \to \delta^{ij}/2c^2$, $S^{ij} \to \delta^{ij}/c^2$, $K^{ij}\log[B/A] = \mathcal{O}(1/|c|)$ and $\mathcal{A}_0 T^{ij} = \mathcal{O}(1/|c|)$. The w integral $\int d^2w\, \log(c^2)/c^2$ is not convergent at $w \to z$, and the phase $e^{ik\cdot w}$ does not help. The source is the last term of N_{3b} (the $\delta^{ij}/(z-w)^2$ term) and the $i\sigma\pi$ term. So Row III as written is not finite by itself: either this region is removed by a subtraction that is not in the file, or the limit $\Lambda \to 0$ was taken where it does not commute with the w integral.

2. **Sign σ.** It is fixed by the $i\epsilon$ of the energy denominators that produced the logarithms. The reduction above assumes that all "-1" inside the logarithms are continued in the same way; if they were not, extra multiples of $2\pi i\, M^{ij}$ would appear, and those do have poles at $A = B$.

3. **Group I, Row II.** The integration momentum is called k, the same symbol as the external momentum in $e^{-ik\cdot(w'-w)}$; one of them must be renamed.

4. **Plus prescriptions.** The closed forms of the longitudinal integrals in Sections 8.1 and 9.1 assume $[1/\xi]_+ f = [f(\xi) - f(0)]/\xi$ on $0 < \xi < 1$ and $[1/(\eta(1+\eta))]_+ f = [f(\eta) - f(0)]/(\eta(1+\eta))$ on $0 < \eta < \infty$. With a different convention only these closed forms change, not the brackets $\mathfrak{B}$, $\mathfrak{B}_V$.

5. **N_{2a} of Group I Row V.** The statement that N_{2a} and $N_{2b\,\rm rest}$ have the same kernel uses the integration limits $\int_{k^+}^0 dq^+$ exactly as written. If the limits are meant as $\int_0^{k^+}$, the $\bar{E}^{im}$ term of N_{2a} has the opposite sign and the two kernels differ by that term.

6. **Distributions.** $\int_0^\infty d\eta\, e^{-i\eta s} = -i/(s - i0)$ in Section 8 is the $\vee \to \infty$ limit in the distributional sense; it includes a $\pi\delta(k\cdot(y'-z))$ piece.

7. **Not reduced.** Group I Row II is left in momentum space. Convergence of the transverse integrals in Sections 6–9 was not examined.